Network Science Formula Cheat Sheet

1. Residue Interaction Graph (RIG)

Symbols:
A_{ij} = adjacency matrix entry (0 or 1)
r_{ij} = Euclidean distance between residues i and j
r_c = cutoff distance (threshold)
$H(\cdot)$ = Heaviside step function: $H(x) = 1$ if $x \geq 0$, else 0

Adjacency (edge exists if within cutoff):

$$A_{ij} = \begin{cases} H(r_c - r_{ij}) & i \neq j \\ 0 & i = j \end{cases}$$

i.e. $A_{ij} = 1$ if $r_{ij} \leq r_c$, else $A_{ij} = 0$.

Typical cutoff range: $5\,\text{Å} \leq r_c \leq 10\,\text{Å}$

Long-range interaction (LRI): residues must be at least 12 apart in sequence:

$$|i - j| \geq 12$$

2. Communities

Symbols:
k_i^{int} = degree of node i counting only edges inside community
k_i^{ext} = degree of node i counting only edges going outside
l_{int} = number of edges entirely inside community C
l_{ext} = number of edges from C to rest of graph
N_c = number of nodes in community C
N = total nodes in graph
δ^{int} = internal density of C
δ^{ext} = external density of C
$\langle k \rangle$ = mean degree of whole graph

Internal density (fraction of possible edges inside):

$$\delta^{\text{int}}(C) = \frac{l_{\text{int}}}{N_c(N_c - 1)/2}$$

External density (fraction of possible cross-edges):

$$\delta^{\text{ext}}(C) = \frac{l_{\text{ext}}}{N_c(N - N_c)}$$

Community conditions (a good community has):

$$\delta^{\text{int}}(C) \gg \langle k \rangle \gg \delta^{\text{ext}}(C)$$

$$k_i^{\text{int}} > k_i^{\text{ext}} \quad \text{(for each node)}$$

$$\textstyle\sum_{i \in C} k_i^{\text{int}} > \sum_{i \in C} k_i^{\text{ext}}$$

3. Similarity & Clustering

Symbols:
$x_{ij}^{(o)}$ = overlap similarity between nodes i and j
$J_n(i,j)$ = number of common neighbors of i and j, plus 1 if i–j edge exists
k_i = degree (number of neighbors) of node i
$S(e_{ik}, e_{jk})$ = Jaccard similarity between two edges sharing node k
$n_+(i)$ = neighbors of i, including i itself

Overlap similarity (how many neighbors i, j share, normalized):

$$x_{ij}^{(o)} = \frac{J_n(i,j)}{\min(k_i,\, k_j)}$$

Link clustering similarity (Jaccard of neighbor sets):

$$S(e_{ik}, e_{jk}) = \frac{|n_+(i) \cap n_+(j)|}{|n_+(i) \cup n_+(j)|}$$

4. Modularity

Symbols:
M = modularity score; $M \in (-1, 1)$; higher = better community structure
L = total number of edges in graph
k_i, k_j = degrees of nodes i and j
P_{ij} = expected number of edges between i and j in random graph
$\delta(C_i - C_j) = 1$ if i, j in same community, else 0
l_c = edges inside community c
k_c = sum of degrees of all nodes in community c

$$M = \frac{1}{2L} \sum_{i,j} (A_{ij} - P_{ij})\, \delta(C_i - C_j)$$

Null model (random graph expected edges):

$$P_{ij} = \frac{k_i k_j}{2L}$$

Equivalent community-sum form:

$$M = \sum_c \left[\frac{l_c}{L} - \left(\frac{k_c}{2L} \right)^2 \right]$$

5. Motifs

Symbols:
Z = Z-score: how significant a motif is vs. random
N_{real} = count of motif in real network
N_{rand} = mean count in randomized networks
SD = standard deviation across random networks
$p_c(k)$ = percolation threshold for k-cliques
k = clique size (number of nodes in clique)
N = number of nodes

Z-score (motif significance):

$$Z = \frac{N_{\text{real}} - N_{\text{rand}}}{\text{SD}}$$

Clique percolation threshold (edge probability needed for giant clique cluster):

$$p_c(k) = \frac{1}{[(k-1)N]^{1/(k-1)}}$$

Special cases:

$$k = 2 \;\Rightarrow\; p_c \sim \tfrac{1}{N} \quad \text{(standard percolation)}$$

$$k = 3 \;\Rightarrow\; p_c = \tfrac{1}{\sqrt{2N}} \quad \text{(triangles)}$$

6. Centrality

Symbols:
$C_B(v)$ = betweenness centrality of node v
σ_{st} = total number of shortest paths from s to t
$\sigma_{st}(v)$ = shortest paths from s to t that pass through v

Betweenness centrality (how often v sits on shortest paths):

$$C_B(v) = \sum_{\substack{s \neq v \neq t \\ s < t}} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

7. Barabási–Albert (BA) Model

Symbols:
t = time step (= number of nodes added)
N = total nodes = t
L = total edges = mt
m = edges added per new node
$\langle k \rangle$ = mean degree = $2m$
$k_i(t)$ = degree of node i at time t
t_i = time step when node i was added
p_k = probability that a random node has degree k
γ = power-law exponent = 3 in BA
D = network diameter
C = clustering coefficient

$$N = t \qquad L = mt$$

$$\langle k \rangle = 2m \qquad k_i(t) = m \left(\tfrac{t}{t_i} \right)^{1/2}$$

Degree distribution is scale-free:

$$p_k \sim k^{-\gamma}, \qquad \gamma = 3$$

Diameter (grows very slowly with N):

$$D \sim \tfrac{\log N}{\log \log N}$$

Clustering (much higher than random graph):

$$C_{\text{rand}} = \tfrac{\langle k \rangle}{N} \sim \tfrac{1}{N}$$

$$C_{\text{BA}} \sim \tfrac{m (\ln N)^2}{8N}$$

8. Scale-Free Networks

Symbols:
p_k = degree distribution: probability a node has degree k
γ = power-law exponent (typically $2 < \gamma < 3$ in real nets)
$\zeta(\gamma)$ = Riemann zeta function (normalization constant)
$k_{\min}$ = minimum degree
$k_{\max}$ = natural cutoff (max degree expected in network of size N)
$\langle k \rangle$ = mean degree
$\langle k^2 \rangle$ = second moment; diverges when $\gamma \leq 3$ (hubs dominate)
$\langle d \rangle$ = average shortest-path distance between nodes

Discrete degree distribution:

$$p_k = \frac{k^{-\gamma}}{\zeta(\gamma)}$$

Continuous (power-law with minimum degree):

$$p(k) = (\gamma - 1)\, k_{\min}^{\gamma - 1}\, k^{-\gamma}$$

Natural degree cutoff (largest hub expected):

$$k_{\max} \sim k_{\min} N^{1/(\gamma - 1)}$$

Moment conditions (what's finite depends on γ):

$$\gamma \leq 2: \quad \langle k \rangle \to \infty \quad \text{(unphysical)}$$

$$2 < \gamma < 3: \quad \langle k \rangle \text{ finite, } \langle k^2 \rangle \to \infty$$

$$\gamma > 3: \quad \langle k \rangle, \langle k^2 \rangle \text{ both finite}$$

Average distance (ultra-small-world behavior for small γ):

$$\langle d \rangle \sim \begin{cases} \text{const} & \gamma = 2 \\ \dfrac{\ln \ln N}{\ln(\gamma - 1)} & 2 < \gamma < 3 \\ \dfrac{\ln N}{\log \log N} & \gamma = 3 \\ \ln N & \gamma > 3 \end{cases}$$

Degree fluctuations:

$$k \approx \langle k \rangle \pm \sqrt{\langle k \rangle} \quad (\gamma > 3\text{: finite spread})$$

$k \approx \langle k \rangle \pm \infty \quad (2 < \gamma \leq 3$: hubs dominate)

$\gamma < 2 \Rightarrow$ non-graphical (impossible degree sequence)

Configuration model (random graph with given degrees):

$$p_{ij} = \frac{k_i k_j}{2L - 1}$$